

Education			
Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (May 2025 – Sept 2025)

SpliDT Project: **Scaling Stateful Decision Tree Algorithms in P4** *P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- Switch-Native Architecture:** Compiled ML decision trees directly into P4 data planes with TCAM-efficient Subtree IDs to resolve hardware ASIC constraints.
- Validation & Optimization:** Built a P4Runtime testing pipeline (Tofino/Mininet) improving reliability by **30%**; hit **10 Gbps** line rate with **20%** lower inference latency.

Open Source Contributions

- Project-HAMi/HAMi:** Fixed scheduler and device-plugin bugs in HAMi, a CNCF project for GPU/device resource sharing and isolation across Kubernetes clusters - **5 PRs** merged.
- WasmEdge/WasmEdge:** **#4470** fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch; **#4466** feat(docker): add standalone alpine-base image build pipeline.
- facebook/bpfilter:** **#507** build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

Cutable | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* (Project Link)

Objective: Generate editable React applications from text and visual requirements.

Approach: Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.

Impact: Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

DMIE | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (Project Link)

Objective: Determine whether a startup idea already exists by comparing it against more than **500,000** company records.

Approach: Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.

Impact: Raised USD **10,000** and serves more than **100** users worldwide.

Moodish | *JavaScript, Node.js, OpenAI API, MCP* (Project Link)

Objective: Rank meal options from a user’s craving, budget, and dietary constraints while keeping the recommendation explainable.

Approach: Built intent extraction and deterministic ranking around an AI-generated summary; added a recommendation trace and confirmation-gated cart preview.

Impact: Produces safer, inspectable food recommendations instead of opaque model output.

Myprod | *Go, Nomad, Traefik, WireGuard, Docker* (Project Link)

Objective: Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

Approach: Built a Go CLI and operator dashboard over Nomad scheduling, Traefik ingress, and WireGuard networking, with guarded node lifecycle controls and live resource metrics.

Impact: Provides one control surface for deploying services across heterogeneous free-tier and temporary compute.

Technical Skills

- Languages:** Python, Go, TypeScript, JavaScript, C++
- AI & Orchestration:** OpenRouter, Azure OpenAI, MCP, E2B, LLM tool-calling, vector search, Sklearn
- Backend & Data:** FastAPI, NodeJS, gRPC, Protobufs, WebSockets, PostgreSQL, Redis, MongoDB
- Tools / Platforms:** Docker, Nomad, CI/CD, Next.js, React, Linux

Positions of Leadership

- Technical Writer, Cloud Native Community Group, New Delhi** (May 2024 – Dec 2024)
 - Led **6+** writers and **4** designers on documentation and event content; social media revamp lifted engagement **60%**.
- Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (Nov 2023 – Nov 2025)
 - Co-led **10+** tech events; anchored GDSC WOW at IIIT Delhi (**800+** live audience, **31+** crew).

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, “Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions,” *2026 ICCSC*, Ghaziabad, India, Feb. **2026**. DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

- 2x** AKTU State TT Gold (**2023, 2024**); Robotics Olympiad Bronze, Thailand (**2019**); NASA, NPTEL & ISRO certifications